Reflective Journal

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Class: 6263-ITAI-1371 Intro to Machine Learning-S10-12321

Module 05 Lab — Data Preparation Reflection (Team ML_1371_12321)

In this lab we worked through the core steps of preparing raw data for a machine learning model, using the Titanic dataset. Our goal was to take data that wasn't model-ready — it had missing values and text columns — and clean it up into a usable form. We focused on three techniques: handling missing values, encoding categorical data, and scaling features.
First, we dealt with the missing values in the 'Age' column. There were 177 missing entries, and we filled them in using the median rather than the mean, since age can be skewed by a few outliers. After imputing, we confirmed there were 0 missing values left. Next, we used one-hot encoding to convert the 'Sex' and 'Embarked' columns from text into numeric columns, since models can't read text directly. Finally, we applied feature scaling to 'Age' and 'Fare' so that the larger 'Fare' numbers wouldn't be treated as more important just because of their size.
What we found most confusing at first was the code itself knowing what to write in each "Your Code Here" section and how the pieces fit together. What surprised us was that once we worked through it, the code actually ran and did what it was supposed to. What clicked for us was seeing how much the Python libraries (like pandas and scikit-learn) do for us a single function like `pd.get_dummies()` or `StandardScaler` handles a whole step that would be really tedious to do by hand, once you know how to use it. We can also see how this connects to the real world: retail companies deal with messy customer and sales data that has missing values and text categories all the time, so the same preparation steps would be needed before they could build a model from it.